Principles of Complex Systems, Vols. 1, 2, & 3D
CSYS/MATH 6701, 6713, & a pretend number
University of Vermont, Fall 2024
"I don't know what I expected"
Solutions to Assignment 03

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Episode links: Wikipedia, IMDB, Fandom, TV Tropes.

Name:

Conspirators:

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1. As in assignment 1, consider a random variable X with a probability distribution given by

$$P(x) = cx^{-\gamma},$$

where c is a normalization constant you determined in the first assignment, and $0 < a \leq x \leq b$. (a and b are the lower and upper cutoffs respectively.) Assume that $\gamma > 1$.

Note: For all answers you obtain for the questions below, replace c by the expression you obtained in the first assignment (below), and simplify expressions as much as possible.

Compute the nth moment of X which is in general defined as:

$$\langle x^n \rangle = \int_a^b x^n P(x)\, \mathrm{d}x$$

Solution:

Begin by substituting P(x) and setup the integral:

$\langle x^n \rangle = \int_a^b x^n \cdot cx^{-\gamma}$

Simplify:

$c \int_a^b x^{n-\gamma}$

Take the integral:

$c \int_a^b \frac{x^{n-\gamma+1}}{n-\gamma+1}$

Evaluate:

$c(\frac{b^{n-\gamma+1} - a^{n-\gamma+1}}{n-\gamma+1})$

Simplify:

if $n - \gamma \neq -1 : \frac{c}{n-\gamma+1}(b^{n-\gamma+1} - a^{n-\gamma+1})$

$\square$

2. **Solution:**

   Here we need to think about 2 cases:

   Case 1: $\gamma > n + 1$ The nth moment is finite and will fluctuate with a and γ.

   Case 2: $\gamma < n + 1$ In this case b will run away, and the moment will grow without bound. There will be too much space under the tail.

   $\square$

3. (a) Find σ, the standard deviation of X for finite a and b, then obtain the limiting form of σ as $b \to \infty$, noting any constraints we must place on γ for the mean and the standard deviation to remain finite as $b \to \infty$.

   Some help: the form of σ^2 as $b \to \infty$ should reduce to

$$= \frac{(\gamma - c_1)}{(\gamma - c_2)(\gamma - c_3)^2} a^2$$

   where c_1, c_2, and c_3 are simple, meaningful constants to be determined (by you).

   (b) For the case of $b \to \infty$, how does σ behave as a function of γ, given the constraints you have already placed on γ? More specifically, how does σ behave as γ reaches the ends of its allowable range?

   Solution:

   For (a) we proceed as such:

   First we find the second moment by substituting n with 2:

   $\langle x^2 \rangle = \frac{c}{3-\gamma}(b^{3-\gamma} - a^{3-\gamma})$

   Now we will substitute $\langle x \rangle$ and $\langle x^2 \rangle$ into $\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2$

   $\langle x^2 \rangle = \frac{c}{3-\gamma}(b^{3-\gamma} - a^{3-\gamma}) - (\frac{c}{n-\gamma+1}(b^{n-\gamma+1} - a^{n-\gamma+1}))^2$

   (b) We have a DWR (a Peterism or technical term which translates to danger Will Robinson):

   $\langle \sigma^2 \rangle \to \infty$ if $\gamma \leq 3$ $\square$

4. Drawing on a Google vocabulary data set:

   (a) Plot the frequency distribution N_k representing how many distinct words appear k times in this particular corpus as a function of k.

(b) Repeat the same plot in log-log space (using base 10, i.e., plot $\log_{10} N_k$ as a function of $\log_{10} k$).

Solution:

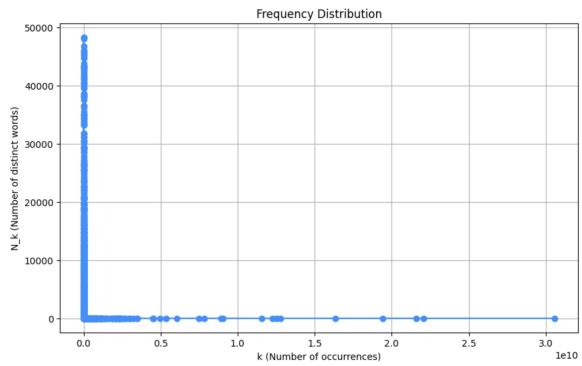


Figure 1: (a) Frequency Distribution

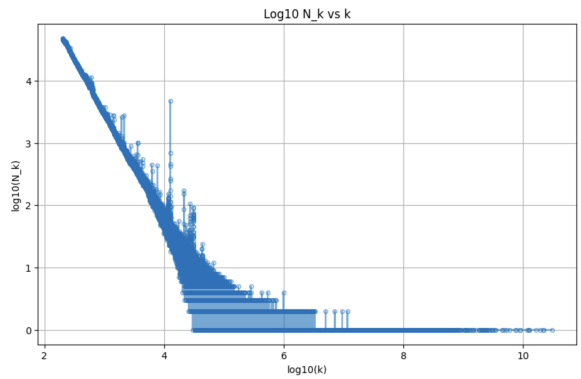
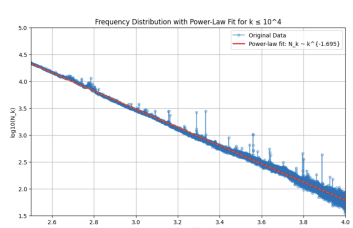


Figure 2: (b) N_k as a function of k

□

5. Using your eyeballs, indicate over what range power-law scaling appears to hold and, estimate, using least squares regression over this range, the exponent in the fit $N_k \sim k^{-\gamma}$ (we'll return to this estimate later).

Solution: □



6. Compute the mean and standard deviation for the entire sample (not just for the restricted range you used in the preceding question). Based on your answers to the following questions and material from the lectures, do these values for the mean and standard deviation make sense given your estimate of γ?

   Hint: note that we calculate the mean and variance from the distribution N_k; a common mistake is to treat the distribution as the set of samples. Another routine misstep is to average numbers in log space (oops!) and to average only over the range of k values you used to estimate γ.

   Solution:

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Mean of the distribution: 61651.635
Standard deviation of the distribution: 16783862.151
Mean with max values from raw_frequency: 75703.776
Standard deviation with max values from raw_frequency: 40228496.004
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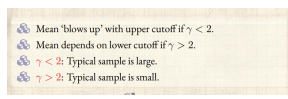
   We can see from the above our mean is influenced heavily by the max values in the raw frequency and given what we see on the slide below from lecture, and the fact that our gamma falls in this range (1.69) we mig



   □

7. (a) A parent has two children, not twins, and one is a girl born on a Tuesday. What's the probability that both children are girls?

   See if you can produce both a calculation of probabilities and a visual explanation with shapes (e.g., discs and pie pieces).

   Once you have the answer, can you improve our intuition here? Why does adding the more detailed piece of information of the Tuesday birth change the probability from $1/3$?

   (Assume 50/50 birth probabilities.)

   (b) Same as the previous question but we now know that one is a girl born on December 31. Again, what's the probability that both are girls?

   Solution:

   The calculation I used can be applied in both. To generalize we should define some variables. We can call GG, GB, BG, BB, which are the possible outcomes, po and we will call Tuesdays or Holidays or whatever constraints or c. For either we can just perform the following:

   $\frac{(.5\cdot(po\cdot c)-1}{(po\cdot c)-1}$

$\frac{(.5\cdot(4*7))-1}{(4\cdot 7)-1} = \frac{13}{27}$

For the December 31 problem we have:

$\frac{(.5\cdot(4*365))-1}{(4\cdot 365)-1} = \frac{729}{1459}$

This calculation actually came out of the process, which I did on paper initially. I just wrote out the possibilities for the Tuesday problem. To visualize that, I have done something terribly unoriginal and created a table as seen below.

GG	GG	GB-G	GB-B	BG-B	BG-G	BB	BB
Tuesday	Monday	Tuesday	Row1	Monday	Tuesday	Monday	Monday
Tuesday	Tuesday	Tuesday	Tuesday	Tuesday	Tuesday	Tuesday	Tuesday
Tuesday	Wednesday	Tuesday	Wednesday	Wednesday	Tuesday	Wednesday	Wednesday
Tuesday	Thursday	Tuesday	Thursday	Thursday	Tuesday	Thursday	Thursday
Tuesday	Friday	Tuesday	Friday	Friday	Tuesday	Friday	Friday
Tuesday	Saturday	Tuesday	Saturday	Saturday	Tuesday	Saturday	Saturday
Tuesday	Sunday	Tuesday	Sunday	Sunday	Tuesday	Sunday	Sunday

Figure 3: Visual Explanation: Tuesday Daughter

$\square$